

TurbineGuard • Engineering Validation Report

Automated Controller Verification Suite | Pass Rate: 100.0% | Date: 2026-10-04 15:20 UTC

Total Requirements	Passed	Failed	Pass Rate	Execution Time
12	12	0	100.0%	68.6 ms

1. Requirements Verification Traceability Matrix

Req ID	Requirement Title	Status	Measured	Target +/- Tol
REQ-001	Rotor Speed Regulation in Region 2	PASSED	1373.40 RPM	1100.00 +/- 400.00
REQ-002	Rated Power Regulation in Region 3	PASSED	2499.80 kW	2500.00 +/- 125.00
REQ-003	Critical Rotor Overspeed Hard Trip	PASSED	50.00 ms	250.00 +/- 50.00
REQ-004	High Wind Cut-Out Storm Shutdown	PASSED	90.00 deg	90.00 +/- 5.00
REQ-005	Generator Thermal Overheat Protective ...	PASSED	50.00 ms	1000.00 +/- 200.00
REQ-006	Thermal High-Warning Power Derating	PASSED	1625.00 kW	1625.00 +/- 50.00
REQ-007	Structural Vibration Trip	PASSED	100.00 ms	300.00 +/- 100.00
REQ-008	Sensor Data Integrity and Failsafe	PASSED	50.00 ms	200.00 +/- 100.00
REQ-009	Communication Bus Loss Protection	PASSED	50.00 ms	500.00 +/- 100.00
REQ-010	Pitch Actuator Normal Rate Limiting	PASSED	8.00 deg/s	8.00 +/- 0.10
REQ-011	Automatic Fault Recovery Hysteresis	PASSED	5.00 s	5.00 +/- 0.50
REQ-012	Extreme Operating Gust (EOG) Dynamic S...	PASSED	1516.96 RPM	1725.00 +/- 0.00

2. Requirements Distribution & Execution Latency

